

# How RURAL BROADBAND Over COGNITIVE RADIO NETWORKS Helps



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**S**tudies conducted by The World Bank show that there is a direct link between a country's gross domestic product (GDP) and broadband penetration. On an average, a ten percentage-point increase in broadband penetration results in a 1.38 percentage-point increase in the GDP of a developing country. Further, studies involving 33 Organisation for Economic Cooperation and Development (OECD) countries show that doubling the broadband speed increases the GDP by 0.3 per cent.

In India, the facilities and conveniences offered by broadband Internet and mobile connectivity have not reached rural pockets, where approximately 70 per cent of the population lives, because telecom network operators do not find it economically viable to do business there. This is due to low population density and high cost of initial investment.

The Indian government has initiated projects to take fibre connectivity till the *gram panchayat* level; a *gram panchayat* being the lowest local self-governing administrative body that governs a group of villages. This still leaves the last-mile connectivity unaddressed.

Providing last-mile connectivity using wired means is not cost-effective—low population density, difficult terrain and insurgency being key factors. Hence, there is a strong need to create wireless broadband technology solutions that can cover large and thinly-populated areas in a cost-efficient manner that businesses find economically-viable and scalable.

## Broadband in the Indian context

Countries define broadband differently and revise the definition as technology advances. As per Telecom Regulatory Authority of India (TRAI), the body that advises the Indian government on policies related to telecommunications, broadband

is defined as "A data connection using any technology that is able to support interactive services including Internet access and support a minimum download speed of 512 kilobits per second (kbps)."

TRAI further recommends "that the upload speed will at least be half the download speed. This definition of broadband (both wireline and wireless) is effective from January 1, 2011. The stipulated download speed of 2Mbps will be effective from January 1, 2015."

While International Telecommunications Union (ITU) classifies broadband as fixed versus mobile, TRAI has merged the definitions of wireline and wireless broadband and included services and speed in its definition on the advice of various stakeholders as well as examining trends in broadband technologies and policies of several countries such as the USA, Germany and Japan.

## TV band spectrum and cognitive radio for broadband

There is an increasing urgency to provide broadband Internet services to rural and underserved communities, both from an economic and social perspective. One of the biggest challenges in providing wireless last-mile connectivity to rural India is cost of spectrum.

Radio spectrum is owned by the government and is auctioned to network operators who prefer to offer services in urban areas. Hence, additional radio bands have to be identified. Since spectrum is allocated in bands, statically and exclusively, to various services on a long-term basis, it has led to most of the spectrum being assigned, leaving very little for additional services.

This prompted the research community to look closely at spectrum utilisation to find out temporal and spatial efficiency of utilisation, leading to the emergence of